Contour-Enhanced Visual State-Space Model for Remote Sensing Image Classification

Liyue Yan, *Student Member, IEEE*, Xing Zhang, *Member, IEEE*, Kafeng Wang, *Member, IEEE*, and Dejin Zhang

***Abstract*— The accurate classification of remote sensing (RS) images can quickly identify various geographical features, which is important for planning, utilizing, and protecting natural resources. Recently, the visual Mamba model, as an extension of the vision transformer (ViT), is attracting widespread attention due to its global receptive field and linear complexity. However, the self-attention mechanism of visual transformers can lead to feature collapse in the deep layers, resulting in the disappearance of low-level visual features. In RS images, low-level features, and especially luminance gradient features, can help discern object boundaries and contour information. This is beneficial for the accurate classification of images but has not been fully leveraged. To make full use of contour information and explore the impact of using handcrafted low-level features on the deep layers of the model, in this study, a contour-enhanced Mamba model based on Vision Mamba (VMamba), is proposed, named G-VMamba. The core novelty of G-VMamba lies in its contour enhancement module (ConEM). First, two separate paths are used to extract adaptive luminance gradients and multidimensional convolutional features at each network layer. Subsequently, the features are combined to impose the constraints of low-level features onto the deeper networks. RS image classification experiments were conducted to evaluate the model's performance, and the results demonstrate the superior performance of G-VMamba in classification tasks. An analysis of class activation maps (CAMs) across different categories shows that G-VMamba focuses more on color (or luminance) change significantly regions in images than models like VMamba, highlighting the efficacy of contour enhancement. The code will be available at: https://github.com/yanliyue/Contour-enhanced-Visual-State-Space-Model.**

***Index Terms*— Feature fusion, image classification, low-level visual features, luminance gradient.**

I. Introduction

SCENE classification of remote sensing (RS) images is a critical technique for rapidly identifying and differentiating among different terrain features, such as forests, croplands, and urban structures [1], [2], [3], [4], and is, therefore, an essential requirement of RS applications such as the planning, utilization, and safeguarding of natural resources. Currently, high-resolution optical RS images achieve meter- or submeter-level spatial resolution, allowing the clear depiction of the spatial structure, surface texture, and edge details of ground objects [5], [6]. Higher resolutions provide not only richer features but also increase the complexity of information, making the extraction and utilization of useful information an important research point for improving the accuracy of RS image classification [7].

Over the past decade or so, deep learning (DL) is being increasingly applied for the data-driven modeling of image representations, achieving great success in image classification or segmentation [8]. Convolutional neural networks (CNNs) and vision transformers (ViTs) are currently two mainstream frameworks of DL, which have been employed for the automation of image feature extraction and representation, and are widely used in image classification tasks [9], [10], [11], [12]. However, they exhibit some limitations in terms of feature utilization. For instance, the convolutional and pooling operations in CNNs tend to discard vital low-level features as the network depth increases [13]; likewise, the ViTs self-attention mechanism can lead to feature collapse in deeper layers [14]. To reduce the feature loss, researchers have explored various strategies to optimize feature exploitation. For example, a classic strategy is to use multiscale convolutions to enhance feature diversity [15], [16], which is widely adopted due to its effective adaptability to objects of different sizes. Another common strategy is to use residual connections to effectively propagate shallow features, mitigating the vanishing gradient problem in deep neural networks and making the training of very deep networks feasible [13]. In CNNs, ResNet alleviates the inherent vanishing gradient issue in CNN structures by utilizing residual blocks, while the more recent ResNeSt [17] incorporates a mechanism called "split-attention," which enhances the model's feature representation capabilities. In ViTs, the Swin Transformer [18] optimizes the classical ViT through shifted windows and a hierarchical design, both of which are integrated within residual connections to facilitate gradient flow in deep networks. Rev-ViT [19] introduces reversible residual blocks, which allow for the retention of all inputs during forward propagation and the computation of gradients through reversible operations during backpropagation, thereby reducing memory usage. Nonetheless, overly deep networks may introduce redundant features, impinging upon the model's generalization ability and feature representation efficiency.

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In the domain of optical RS imagery, variations in scenes commonly manifest as differences in land use categories and their spatial distributions. When analyzed at the image level, these disparities become prominently evident through shifts in color, texture, and luminosity. A critical feature describing the degree and direction of such color or luminosity changes is the image gradient, which represents a quantified measure of the rate of change in pixel intensity, thereby facilitating the identification of significant features, such as object boundaries, textures, and orientations [20], [21], [22]. High-resolution RS imagery contains a rich array of visual characteristics and elaborate spatial associations. Extracting low-level features, particularly contour information, from these images is of significant importance for distinguishing between various types of ground surfaces. The absence or loss of such low-level features can impair a model's ability to recognize fine-grained details, such as sparsely distributed buildings, potentially leading to reduced accuracy in classification tasks.

An increasing number of researchers have directed their attention to the significance of gradient features in images. In the past two years, contour-aware and edge-enhanced attention schemes have been introduced [23], [24], [25], aiming to enhance the model performance through an increased focus on edge contours. However, this reliance on additional attention modules fails to transcend the spatial limitations imposed by convolutional kernels, leading to inferior model generalization ability and a substantial increase in computational complexity. Thus, some researchers, considering that traditional edge detection techniques (such as the Canny or Sobel operator) excel at identifying object boundaries and contours with high computational efficiency and strong interpretability [26], [27], [28], have attempted to enhance DL performance using conventional methods such as gradient or edge. One approach involves preprocessing images with edge detection algorithms, then concatenating the generated edge maps as additional input channels to deep neural networks [29]. Another method entails designing dedicated layers within the network to extract edge features, enabling it to identify and leverage edge information during the learning process autonomously; this has proven effective in enhancing the accuracy of image classification [30], [31], [32]. However, these strategies also present challenges, including increased computational demands and reduced applicability to images with scarce or ambiguous edge information. It is evident that two key issues currently persist in RS image classification.

1) How can the loss of low-level features in deeper layers of the network be mitigated without increasing the computational load incurred by the model?
2) How can gradient features be perceived effectively and the contribution of features balanced to accommodate diverse ground imaging scenarios?

Recently, Vision Mamba (VMamba), as an extension of the ViT, attracted widespread attention due to its simultaneous global receptive field and linear complexity [33]. As a visual state-space (VSS) model, VMamba is an adaptation of the original Mamba architecture, which was designed for 1-D causal sequences, to suit 2-D noncausal images. In this article,

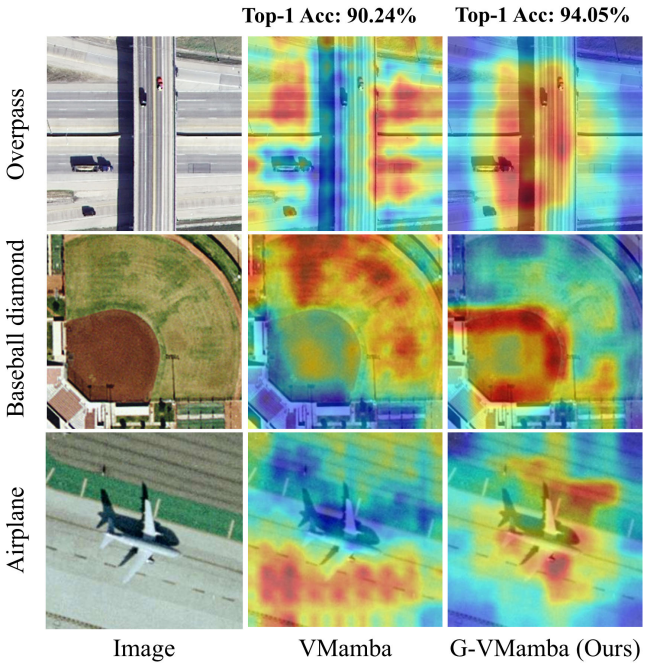


Fig. 1. CAM visualization for VMamba [33] and G-VMamba models' classification of UC-Merced dataset images. When the model classifies the scenes in the image, the G-VMamba model focuses on areas where the color (or brightness) of the image changes more significantly (red areas), such as the lane intersection position of the overpass scene, the edge of the court in the baseball diamond scene, and the airplane shadow and lawn border of the airplane scene.

we propose an innovative approach based on VMamba, which combines hand-crafted image gradients with DL's convolutional features, leading to the establishment of the G-VMamba model. The core novelty of the G-VMamba model resides in the integration of a contour enhancement module (ConEM) into VMamba's forward propagation process, which integrates hand-crafted features with the strengths of DL, thereby bolstering the model's sensitivity to variations in color or luminance (Fig. 1). The G-VMamba model incorporates two distinct computational paths designed to handle adaptive gradient information and multidimensional convolutional features separately. The features output by these two paths are fused at each network layer, thereby reinforcing the contour information content of features across every network layer without altering the model's complexity. Finally, the performance of the model is demonstrated through a scene classification experiment of high-resolution RS images. Further analysis of the features shows the specific impact of ConEM's use of manual low-level features on the model.

The main contributions of this article are as follows.

1) We propose an enhanced Mamba framework for low-level feature augmentation, named G-VMamba, which effectively enhances the low-level features in each network layer through scale-adaptive hand-crafted gradient operations, thereby enhancing the model's classification accuracy.
2) We devised a ConEM that combines hand-crafted features with DL features using a feature separation and fusion mechanism. This improves the model's sensitivity to features related to brightness changes while retaining

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its ability to perceive mid-level features such as texture and high-dimensional features.

3) Our extensive experiments on three RS datasets confirm that the proposed G-VMamba outperforms existing models, especially on smaller datasets.

II. Related Works

A. DL-Based RS Image Classification

The evolution of DL has enabled the automation of image feature extraction and representation, allowing data-driven methods to achieve effective image representations, resulting in significant breakthroughs in classification tasks. However, DL typically employs fixed-size convolutional kernels, which can lead to a loss of useful information and poor adaptability to varying scenes. Consequently, innovative approaches, such as U-Net, PSPNet, and deformable convolution have been proposed [16], [34]. Among these, U-Net [35] leverages skip connections to learn spatial correlations across different levels, and in RS image classification tasks, U-Net's skip connection mechanism aids in preserving spatial information, making it suitable for extracting the fine-grained features necessary for classification. PSPNet [15] introduced the pyramid pooling module, which aggregates global context information at multiple scales, enhancing the network's capability to handle objects of varying sizes and shapes, thereby lending itself well to a broad range of RS image classification tasks with rich multiscale features. Nonetheless, as the network depth increases, the convolutional and pooling operations in CNNs tend to discard crucial low-level features. Consequently, ResNet [13], featuring residual structures, was proposed to address the issue of vanishing gradients, and promptly evolved into a fundamental architecture for RS image classification. Nevertheless, excessively deep networks risk incorporating redundant features, which may increase the computational cost of model training and also undermine the model's generalization ability [36]. High-resolution RS images often come in large data sizes and require more efficient models.

ViTs equipped with self-attention mechanisms and optimized architectures were introduced recently and have gained widespread adoption [11], [12], [37]. ViTs segment images into fixed-sized patches and process these patches using self-attention mechanisms, thus bypassing the conventional CNN's limitation of spatially limited local receptive fields, which enables significant improvements for large-scale classification tasks. A representative method is that of Liu et al. [18], who designed the Swin transformer, which incorporates sliding window operations and hierarchical designs, restricting attention calculations within each window while enhancing information exchange between different windows through shifted window attention, thereby improving the computational efficiency and performance of ViT models. However, self-attention mechanisms can lead to feature collapse in deeper layers, resulting in the disappearance of low-level visual features. Some researchers have made advances through the integration of ViTs with U-shaped CNNs and bolstering features through hierarchical connections [38], [39]. To address the issue of feature collapse, Diko et al. [14] introduced residual connections into ViTs to propagate shallow features to deeper networks. However, due to the contrast between the way the residual modules convey shallow features and the global nature of attention mechanisms, their improvement effects are relatively limited within multiscale architectures that rely on local perception.

Considering the utility of object contours and shapes in the accurate depiction and recognition of image elements, some researchers have attempted to enhance image gradient features to boost model performance. Particularly in the domain of medical imaging, contour-aware attention mechanisms and edge-augmented attention schemes have been introduced, aiming to enhance model performance through a focus of the model's attention on edge contours [23], [24], [25]. Nonetheless, these attention modules are often constrained by the inherent spatial dimensions of convolutional kernels, which limit the model's generalization across different scales and complexities of images and significantly increase computational complexity. In RS images, different surface types have different scales and forms. For example, rivers and roads may be distributed linearly throughout the image, sparse buildings tend to have small areas in images, and farmland often appears as a large area covering the entire image, etc. Low-level features, especially contour information, are important for distinguishing different forms but have not been adequately considered.

B. Combine Traditional Gradient Features and DL

In the context of optical RS imagery, distinguishing between various terrain features, such as crops, forests, and urban structures, is contingent on the extraction of the nuanced details of brightness or color variations these image elements exhibit [1], [40], [41]. Characteristics like the transition edges between shaded and illuminated areas and regular or irregular spatial arrangements are pivotal for the differentiation between these areas. A key feature in computer vision that articulates these details is the image gradient, which enables the recognition of surfaces' smoothness or roughness through the calculation of the gradient's magnitude [42], [43], [44]. For instance, water bodies exhibit lower gradients than land, indicating nearly uniform, smooth surfaces, while the more intricate textures of land are generally reflected through higher gradient variations due to factors such as soil particles and vegetation. Consequently, low-level features such as image gradients form the bedrock of understanding high-resolution RS imagery, whether at the macroscopic level of terrain classification or the microscopic analysis of edges and texture detail.

DL can automatically extract abstract features, offering unparalleled advantages over traditional methods. However, manually extracted low-level features retain more crucial details necessary for distinguishing targets and offer greater interpretability [45], [46], [47], [48]. Some researchers have attempted to enhance the performance of DL by incorporating traditional methods, such as gradient operators or edge detection techniques [32], [49], [50]. One approach involves preprocessing images with edge detection algorithms, then

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concatenating the generated edge maps as additional input channels to deep neural networks [29]. Although this approach has the potential to enhance the discriminative power of DL models, it is not universally applicable to all types of images, especially those with less or blurred edge information. In order to address the limitations of these simple preprocessing methods, the hybrid DeepCervix framework, which combines edge detection and DL features, was proposed [51], which balances the contribution of edge features and deep features through parameter adjustment. However, if the dataset is not large enough to accommodate the additional complexity introduced by the hybrid feature set, there is a risk of overfitting. In another example, DeepEdgeNet [30] utilized a dedicated layer to extract edge features, enabling the network to autonomously identify and utilize edge information during the learning process, thereby improving classification accuracy. However, due to the increased computational requirements caused by the increased complexity of the edge perception layer, high-resolution RS images require substantial computational resources that are not always available, so more efficient models are required.

It can be seen that the complementary strengths of low-level visual features and DL models warrant further exploration.

C. Mamba Models

Recently, state-space models (SSMs) [52] have gained significant attention in both the natural language processing and the computer vision domains due to their success in enhancing computational efficiency for long sequences while maintaining comparable performance to ViTs [33], [53]. Mamba is a selective structural SSM that excels in long-sequence modeling tasks, as it alleviates the modeling constraints of CNNs through global receptive fields and dynamic weighting, offering advanced modeling capabilities akin to those of transformers. Consequently, a generic visual representation foundation model (VMamba) has been proposed, which has the global awareness advantage of ViTs and has only linear computational complexity [33]. VMamba has been tested on standard natural scene datasets such as ImageNet-1K, COCO, and ADE20K, demonstrating promising accuracy and computational efficiency. For RS imaging applications specifically, researchers have also initiated explorations with the Mamba model. For instance, Samba [54] was the first to apply SSMs to semantic segmentation of RS images, outperforming methods based on CNNs and ViTs. Meanwhile, RSMamba [55] introduced, for the first time, a dynamic multipath activation mechanism on the top of the Mamba structure for RS image classification tasks, exhibiting good performance across multiple RS image classification datasets.

Nonetheless, from an implementation standpoint, the original design of the Mamba architecture, which was aimed at 1-D causal sequences, poses challenges when adapting to 2-D noncausal images. This has led to the weaker performance of current Mamba-based visual models in local feature extraction compared to CNNs and ViTs. To alleviate this issue, some researchers have proposed combining CNNs with SSMs [33], [56], [57], achieving notable improvements. Taking [33] as an example, by integrating CNN operations within the VMamba framework, local feature extraction can be significantly improved, complementing the global context modeling provided by the SSMs; at the same time, the 2-D selective scanning (SS2D) module is introduced for traversing the 2-D space of an image using four different scanning paths (referred to as cross-scan). This ensures that each pixel in an image can integrate information from different directions, facilitating the establishment of global receptive fields.

Although the introduction of the SS2D module helps mitigate some of these limitations, it might not fully compensate for the weaker local feature perception inherent in SSMs. Therefore, in complex image tasks, VMamba may still face challenges in fine feature extraction and perception. Further integration or optimization of CNN modules could potentially enhance VMamba's performance in these areas.

III. Method

In this section, the overall structure of the proposed G-VMamba is first introduced and the ConEM's function is analyzed. Then, the AG feature module in G-VMamba is described.

A. Network Structure

The overall architecture of the proposed G-VMamba network is shown in Fig. 2.

As presented in [33], VMamba comes in three sizes: tiny (VMamba-T), small (VMamba-S), and base (VMamba-B), which have a four-stage network structure and different numbers of parameters. VMamba-T processes images through four stages consisting of VSS blocks. SSMs, which are a key characteristic of VMamba, model complex data transformations using a set of equations that involve both observable outputs and unobservable system states. SSMs are commonly considered as linear time-invariant systems that map stimulation $x(t) \in \mathbb{R}^L$ to response $y(t) \in \mathbb{R}^L$. These models are formulated as linear ordinary differential equations (ODEs) (1), where the parameters include $A \in \mathbb{C}^{N\times N}$, $B, C \in \mathbb{C}^N$ for a state size N, and the skip connection $D \in \mathbb{C}^1$. The specific defining calculations of parameters A, B, C, and D, refer to the calculations within the S6 block in [33]

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) + Dx(t). \end{aligned} \tag{1}$$

Taking VMamba-T as an example, a given RS input image $X \in \mathbb{R}^{(H\times W\times 3)}$ is processed using four successive VSS blocks. The process is initiated with a division of the image into patches using a stem module, akin to ViTs, but the patches' 2-D form is retained and there is no conversion into a 1-D sequence. The four blocks result in feature maps with varying dimensionalities, denoted by $F_1 \in \mathbb{R}^{((H/4)\times(W/4)\times C_1)}$, $F_2 \in \mathbb{R}^{((H/8)\times(W/8)\times C_2)}$, $F_3 \in \mathbb{R}^{((H/16)\times(W/16)\times C_3)}$, and $F_4 \in \mathbb{R}^{((H/32)\times(W/32)\times C_4)}$.

As depicted in Fig. 2(a), for the forward pass of the VMamba model, hand-crafted gradient features are calculated and integrated with the input data, named the G-VMamba

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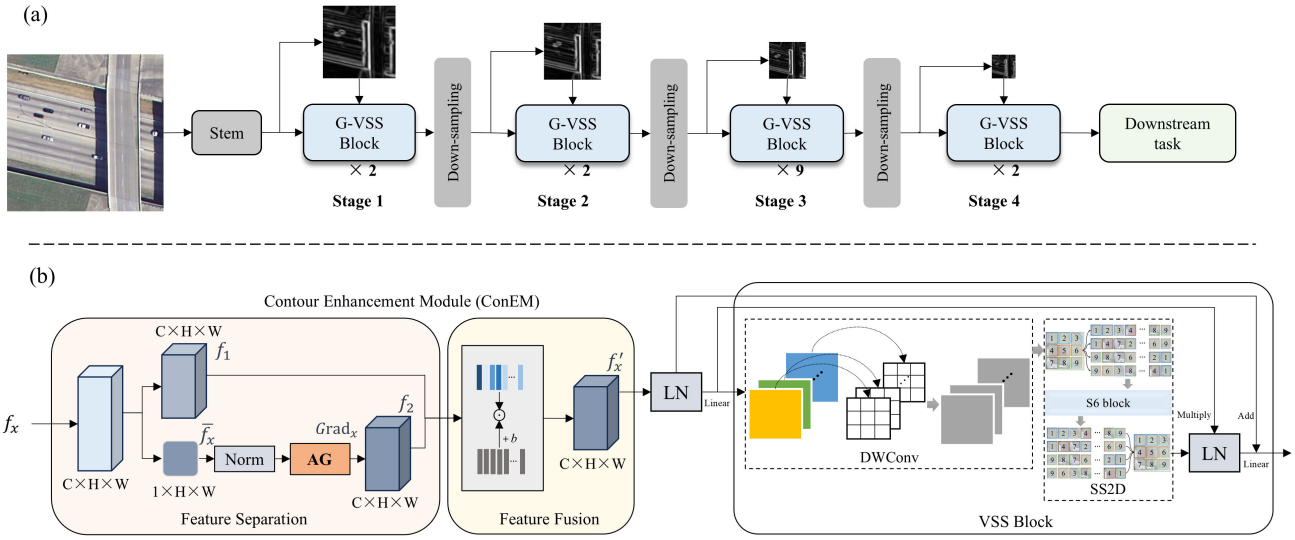


Fig. 2. (a) Overview of G-VMamba model (architecture). (b) Feature grouping in the G-VSS block.

model. The four stages of G-VMamba correspond to four different feature scales. We improve the feature richness by integrating hand-crafted gradient features into these four stages, thereby alleviating the feature collapse problem. At the same time, this operation can enhance the model's ability to perceive brightness changes and improve model performance. The downstream task depicted on the right in Fig. 2(a) is a conventional component employed for image classification and can potentially be replaced with components for other tasks, such as image segmentation or object detection. In Fig. 2(b), the core component of the VMamba model includes a Transformer block using residual connections to alleviate feature collapse, and our G-VMamba continues this design.

As shown in Fig. 2(b), the G-VMamba model's novelty is the ConEM. In the feature separation phase, two paths are employed to separately handle the adaptively calculated luminance gradients f_2 and the multidimensional convolutional features f_1. Following this, a feature fusion strategy is employed to merge these two sets of features at each block. Within the feature separation module, a luminance calculation branch is established, which employs an AG algorithm to individually calculate the luminance gradient features f_2 at every block, ensuring the comprehensive preservation of the luminance gradient, which is a vital low-level feature, in the deeper network layers. Subsequently (Fig. 2(b), middle), dedicated feature fusion steps are tailored for f_1 and f_2, to consolidate their information. The VMamba model's VSS Blocks (Fig. 2(b), right) combined with the ConEM form the model's forward propagation calculation pipeline. In the VSS block, SS2D is employed, which is a method based on a cross-scan mechanism to traverse the image. The four resulting features are then individually processed through the S6 block, and the four output features are merged (scan merge) to construct the final 2-D feature map [33]. The SS2D module is designed to bridge the gap between the effectiveness of 1-D selective scanning for sequential data and the nonsequential nature of 2-D vision data.

B. Contour Enhancement Module

This section is dedicated to elaborating on the proposed contour enhancement method, focusing on the feature separation and fusion processes involved.

1) Feature Separation: Luminance gradient features capture and such edges and directional changes in a quantifiable manner, playing a pivotal role in visual analysis. Consequently, two separate paths are established for luminance gradient features and multidimensional convolutional features (Fig. 2(b), left).

During feature separation, the multidimensional convolutional features f_1 are passed unchanged from the original input feature f_x. The specific method for obtaining the luminance gradient features is as follows: the channelwise mean of the input feature $f_x^{(h\times w\times c)}$, yielding $\overline{f_x}^{(h\times w\times 1)}$. This is then normalized to ensure the feature values of $\overline{f_x}^{(h\times w\times 1)}$ fall within the range $[0, 1]$. Finally, the AG algorithm is employed to calculate the luminance gradient Grad_x from $\overline{f_x}^{(h\times w\times 1)}$, with the gradient values also constrained to the interval $[0, 1]$. To facilitate feature fusion, the channels of Grad_x are replicated to match the original number of channels c of the input feature and are denoted by f_2.

Notably, while the multidimensional features extracted by the network's convolutional operations encompass contours or edges, the edge features extracted using convolutional kernels can sometimes miss crucial details, with this loss becoming more severe in deeper network layers. The AG algorithm calculates luminance changes, which preserves the integrity of contour information across different feature layers (Fig. 3), thus overcoming this limitation. In Fig. 3, $\overline{f_x}$ can be represented using the following equation:

$$\overline{f_x}(h, w) = \frac{1}{c}\sum_{k=1}^{c} f_x(h, w, k). \tag{2}$$

2) Feature Fusion: A straightforward fusion method is employed to integrate the feature information generated by the two separate paths effectively (depicted in Fig. 2(b), middle). The final fused feature output by this module, denoted by f_x',

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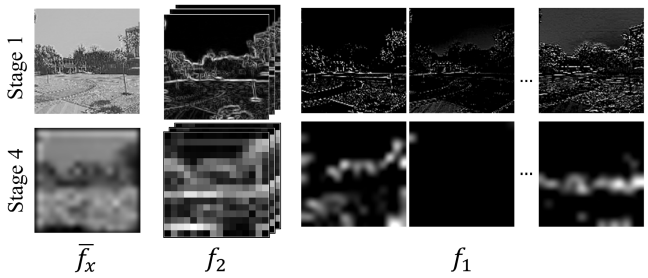


Fig. 3. Comparison of luminance variation features between the feature separation module and those extracted using network convolutions.

is calculated as follows:

$$f_x' = f_1 \odot (f_2 + b) \tag{3}$$

where f_1 represents the multidimensional convolutional features. The luminance gradient feature, $f_2(h, w, c) \in [0, 1]$ has values near 1 at locations of prominent color (or luminance) change and tends to 0 where no such change occurs. To ensure that regions where f_2 approaches 0 do not disrupt color, texture, and other regional characteristics of f_1 during multiplication, a constant matrix b is added to f_2. The shape of b is also (h, w, c). When $b(h, w, c) = 1$, the resulting $(f_2 + b) \in [1, 2]$ is elementwise multiplied with f_1. This operation maintains the color texture features extracted by convolution virtually unchanged while enhancing brightness gradient features by a proportional factor up to a maximum of doubling of their original intensities.

This strategy effectively enhances luminance gradient features on the input features during the model's forward propagation. Hand-crafted adaptive luminance gradient features are integrated at each model layer, steering its attention toward prominent color (or luminance) changes throughout the learning process. The dimensions of the fused features remain consistent with those of the input feature f_x, so the computational complexity of the model is not increased.

To further investigate the impact of incorporating hand-crafted gradient features on model performance, the value of b was varied and the resultant shifts in model performance were observed.

C. Adaptive Gradient Algorithm

In this section, a scale-adaptive gradient computation approach is proposed. The AG algorithm integrates multi-scale Sobel operators and the histogram of oriented gradients (HOGs) operator to extract local gradients and orientations, which are combined to create global gradient awareness.

The Sobel operator reveals locations of abrupt luminance changes such as edges or contours in an image by computing the first-order derivatives (gradients) in the horizontal and vertical directions [26]. In order to maintain a robust representation of luminance gradients across various scales, the kernel size k of the Sobel operator within the AG algorithm is determined through the four stages of the features (Fig. 4).

The equation for calculating the gradient of an image in direction $\theta \in [-90°, 90°]$ using the Sobel operator is as

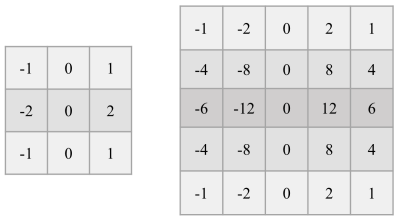


Fig. 4. Sobel operator's horizontal gradient convolutional kernel, K_x. (a) $k = 3$, corresponding to $h < 64$ in $\overline{f_x}$. (b) $k = 5$, corresponding to $h \geq 64$ in $\overline{f_x}$. The vertical gradient convolutional kernel, K_y, is the transpose of K_x.

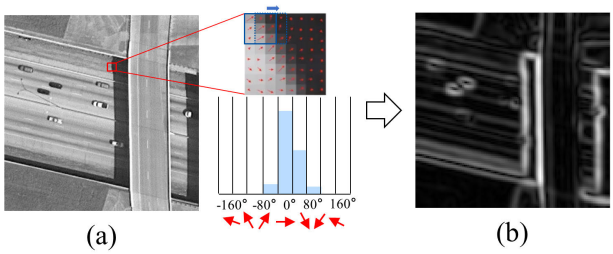


Fig. 5. Example of adaptive Sobel and HOG feature at stage 1. (a) Channel-wise mean $\overline{f_x}$ of the input feature map $f_x^{(h\times w\times c)}$ for depthwise convolution. (b) Luminance gradient feature Grad_x.

follows:

$$\begin{cases} G_y = \sum_{i=-1}^{1}\sum_{j=-1}^{1} I(x+i, y+j)\cdot K_y(i,j) \\ G_x = \sum_{i=-1}^{1}\sum_{j=-1}^{1} I(x+i, y+j)\cdot K_x(i,j) \end{cases} \tag{4}$$

$$\begin{cases} G_\theta = \sqrt{G_x^2 + G_y^2} \\ \theta = \arctan\left(\dfrac{G_y}{G_x}\right) \end{cases} \tag{5}$$

where $I(x, y)$ represents the pixel value of the original image at coordinates (x, y), and $K_x(i, j)$ and $K_y(i, j)$ are the convolutional kernels of size k for the Sobel operator in the horizontal and vertical directions, respectively. G_x and G_y denote the gradients of the pixels in the horizontal and vertical directions. G_θ and θ represent the magnitude and direction of the gradient for each pixel, respectively.

Once G_θ and θ have been calculated, the HOG feature descriptor can be obtained [43]. Local neighborhoods (3×3 cell units) of pixel gradients G_θ and orientations θ are extracted from the image, and these gradients are clustered into predefined direction interval histograms according to their directions. Subsequently, the direction with the highest histogram count is assigned as the gradient direction for the target pixel, with the maximum value in the corresponding direction interval serving as the gradient magnitude. Through this process, a robust gradient feature description is ultimately formed, exhibiting strong resilience and discriminatory power (Fig. 5).

The detailed implementation of the AG algorithm is presented in Algorithm 1. It is worth mentioning that the AG algorithm simultaneously extracts features of both gradient magnitude and orientation; however, the orientation features are utilized solely for generating global gradient features,

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Algorithm 1 Adaptive Gradient

Input: Feature map $\overline{f_x}$, Direction interval $d = 8$.
Output: Gradient feature $Grad_x$.
Parameters: Kernel size k and kernel K of Sobel, the horizontal gradients G_x and vertical gradients G_y, the magnitude G_θ and angle θ of the Sobel.

Get k and the corresponding K by Fig. (4)
Get G_x and G_y by Equ. (4)
Get G_θ and θ by Equ. (5)
Divide $[0°, 180°]$ into d intervals, with id $\in 0, 1, \ldots, d$
Match the corresponding interval id for θ
Initialize $Grad_x$ with the same shape as $\overline{f_x}$
for $i = 0$ **to** d **do**
    Extract the 3×3 neighborhood of G_θ centered where id $= i$, labeled as G_θ^n
    Update $Grad_x$ with the maximum value of G_θ^n where id $= i$
end for
Normalize $Grad_x$
return Gradient feature $Grad_x$

with the gradient direction not being considered in subsequent stages such as feature matching and training.

IV. Experiments

A. Datasets

To evaluate the efficacy of the proposed method, extensive experiments were performed on three RS datasets: UC-Merced [58], NWPU-RESISC45 [59], and AID [60]. Each dataset encompassed a unique assortment of categories and image quantities.

1) UC-Merced [58]: UC Merced is a 21-class land-use RS image dataset, with 100 images per class, each measuring 256×256 pixels. The images were extracted from larger images from the USGS National Map Urban Area Imagery collection for various urban areas around the country. The pixel resolution of this public domain imagery is 0.3 m.

2) NWPU-RESISC45 [59]: This dataset is an open benchmark for RS image scene classification and was created by Northwestern Polytechnical University. It contains 31 500 images of 256×256 pixels divided into 45 scene classes, with each class containing 700 images. Among its notable features, RESISC45 contains varying spatial resolutions ranging from 0.2 m/pixels to more than 0.3 m/pixels.

3) AID [60]: The aerial image dataset is a large-scale dataset of aerial images sourced from Google Earth. It comprises 10 000 labeled scene images divided into 30 scene classes with approximately 200–420 images per class, each measuring 600×600 pixels. Images within the AID are inherently multisource, as the Google Earth images originate from various remote imaging sensors.

B. Implementation Details

To train the proposed model, the dataset was split randomly, with 80% of images allocated for training and 20% for testing. To expedite model convergence, normalization (mean, std) was applied for the per-channel normalization of images, thus enhancing data uniformity and training efficiency.

All experiments were implemented using PyTorch on five NVIDIA A100-SXM4-40 GB GPUs. Following the configuration in [33], the batch size for training was set at 10 (base model), 12 (small), and 24 (tiny), and the training process spanned 300 epochs (with the first 20 epochs acting as a warm-up). The training process incorporates the AdamW optimizer with betas set to (0.9, 0.999), a momentum of 0.9, a cosine decay learning rate scheduler, an initial learning rate of 1×10^{-3}, and a weight decay of 0.05. Additional techniques, such as label smoothing (0.1) and exponential moving average, were also employed. All models were trained using the cross-entropy loss function (6), with the performance metric being top-1 accuracy

$$L = \frac{1}{N}\sum_i L_i = -\frac{1}{N}\sum_i \sum_{c=1}^{M} y_{ic} \log(p_{ic}). \tag{6}$$

Here, L represents the cross-entropy loss; M denotes the total number of classes in the dataset; and y_{ic} is the indicator function (0 or 1), which equals 1 if the true class of the sample i is c, and 0 otherwise. p_{ic} is the predicted probability that sample i belongs to the class c.

C. Performance Comparison

1) Performance Comparison of State-of-the-Art Models: The performance of the proposed G-VMamba was benchmarked against six representative state-of-the-art models from three prominent architectures, namely CNNs (ResNet [13], ResNeSt [17]), ViTs (Swin Transformer [18] and Rev-ViT [19]) and Mamba (RSMamba [55] and VMamba [33]), ResNet alleviates the gradient vanishing problem inherent in CNN structures using residual blocks, while the more recent ResNeSt incorporates a mechanism known as "split attention" which enhances the model's feature representation capabilities. The Swin Transformer and Rev-ViT are notable representatives of the ViT architecture, with the former optimizing ViTs' efficiency and performance via mobile windows and a hierarchical design, while the latter design reduces computational and memory overheads through a reversible architecture. Both models were designed to enhance Transformer efficiency. VMamba serves as the baseline method of this article, with RSMamba being the first application of the Mamba structure in RS image classification. In addition, the RS image classification performance of VMamba-T/S/B models and further discussed the advantages and performance enhancements brought about by the proposed G-VMamba over them.

The open-source project named MMPreTrain [61] was utilized to realize the non-Mamba models. Before each training session, the data configurations and model settings were adjusted according to the specifics of different datasets. The comparison of the classification performance of all methods across various datasets is presented in Table I.

The experimental results reveal the following. Compared with VMamba, G-VMamba exhibits improved classification performance across all scales (tiny/small/base). The performance of the VMamba and G-VMamba models was improved

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TABLE I
TOP-1 ACCURACY COMPARISON ACROSS VARIOUS MODELS ON THE UC-MERCED, NWPU-RESISC45, AND AID DATASETS

Model	Params (M)	UC-Merced (%)	NWPU-RESISC45 (%)	AID (%)
Swin Transformer-S [18]	46.59	92.02	92.21	93.01
Rev-ViT-S [19]	20.68	90.00	89.77	91.03
ResNet 50 [13]	22.46	91.76	91.28	91.37
ResNeSt 50 [17]	24.30	89.98	91.33	90.66
ResNeSt 101 [17]	46.04	90.16	92.45	93.00
RSMamba-L [55]	15.49	91.90	93.33	94.28
VMamba-T [33]	21.90	89.29	94.67	93.97
VMamba-S [33]	43.21	90.24	95.95	93.46
VMamba-B [33]	74.51	91.43	96.43	95.48
G-VMamba-T	21.83	93.57	95.52	95.71
G-VMamba-S	42.36	94.05	96.49	96.40
G-VMamba-B	71.76	95.24	96.89	96.45

TABLE II
TRAINING AND INFERENCE TIMES OF MODELS

Dataset	Model	Scale	Training	Inference	Inference per Image
UC-Merced	VMamba	Tiny	4.80 h	5 s	11.90 ms
		Small	4.92 h	6 s	14.29 ms
		Base	5.40 h	6 s	14.29 ms
	G-VMamba	Tiny	**4.49 h**	**3 s**	**7.14 ms**
		Small	**4.71 h**	**5 s**	**11.90 ms**
		Base	**5.48 h**	**6 s**	**14.29 ms**
NWPU-RESISC45	VMamba	Tiny	23.83 h	17 s	2.7 ms
		Small	31.12 h	36 s	5.7 ms
		Base	40.20 h	44 s	7.0 ms
	G-VMamba	Tiny	**29.80 h**	**21 s**	**3.3 ms**
		Small	**39.02 h**	**42 s**	**6.7 ms**
		Base	**73.62 h**	**63 s**	**10.0 ms**
AID	VMamba	Tiny	10.52 h	11 s	5.5 ms
		Small	13.20 h	12 s	6.0 ms
		Base	21.45 h	19 s	9.5 ms
	G-VMamba	Tiny	**12.62 h**	**12 s**	**6.0 ms**
		Small	**15.00 h**	**15 s**	**7.4 ms**
		Base	**24.08 h**	**22 s**	**11.0 ms**

as the size of the datasets increased. As the model parameters increase, there is a slight improvement in model performance.

The comparison of classification accuracies across different datasets and model sizes revealed two findings. On one hand, the advantage of G-VMamba over the VMamba models became more pronounced as the dataset size decreased. On the UC-Merced dataset, which contained 2100 images, G-VMamba-T surpassed VMamba-T by 4.28 percentage points (pp) in accuracy, G-VMamba-S outperformed VMamba-S by 3.81 pp, and G-VMamba-B's performance was improved over that of VMamba-B by 3.81 pp. It can be seen that G-VMamba's advantage was slightly clearer for smaller model scales. On the NWPU-RESISC45 and AID datasets (with 31 500 and 10 000 images, respectively), G-VMamba-S and G-VMamba-T achieved an increase of approximately 1 pp in accuracy compared with their VMamba counterparts of the same scale, while the difference between G-VMamba-B and VMamba-B was marginal, only increasing by 0.4 and 0.05 pp.

Our experiments also included comparisons of the classification efficacy between CNN models and ViT models, wherein, under equivalent parameter counts, the Mamba models exhibited superior performance. However, the number of parameters of RSMamba is substantially smaller (about 75%) than that of the smallest VMamba and G-VMamba models, while RSMamba's performance metrics are excellent and close to those of VMamba.

2) Comprehensive Comparison of VMamba and G-VMamba: In Table II, we calculate the training and inference time of VMamba and G-VMamba to evaluate the computational efficiency. Overall, the test efficiency of our proposed G-VMamba is lower than that of VMamba. The main reason for the efficiency difference is the ConEM. Only when the model parameters are small and the number of images is small (420 test images of UC-Merced), the efficiency of the two is almost the same.

In Table III, we computed the prediction accuracies of the VMamba and G-VMamba models across different image scenarios and subsequently calculated the mean and standard deviation (SD) of these accuracies.

In the experiment using the UC-Merced dataset, both G-VMamba-T and G-VMamba-S demonstrated strong average classification performance. In contrast, G-VMamba-B exhibited a low average classification score coupled with a high SD, indicating some performance variability across classes. This suggests that, with a smaller dataset (2100 images), even models with a large number of parameters find it challenging to learn sufficient knowledge to balance performance across different classes.

In the experiment with the NWPU-RESISC45 dataset, G-VMamba's average classification score and SD were both poor. This dataset contains more classes (45) than UC-Merced (21) and AID (30), leading us to speculate that the enhancement of brightness gradients may not positively impact all scene classi-

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TABLE III
COMPARISON OF ACCURACY MEAN AND SD ACROSS CLASSES

Dataset	Model	Scale	Mean (%)	SD (%)
UC-Merced	VMamba	Tiny	88.60	10.62
		Small	86.50	9.70
		Base	88.81	7.79
	G-VMamba	Tiny	91.19	10.90
		Small	91.70	9.57
		Base	86.13	14.57
NWPU-RESISC45	VMamba	Tiny	94.55	5.27
		Small	94.69	5.14
		Base	96.40	4.18
	G-VMamba	Tiny	87.85	8.06
		Small	88.44	7.40
		Base	93.23	4.26
AID	VMamba	Tiny	92.20	5.80
		Small	92.41	6.10
		Base	93.12	5.02
	G-VMamba	Tiny	93.92	6.77
		Small	94.00	6.63
		Base	94.05	5.68

TABLE IV
FEATURE LAYER DESCRIPTION FOR CAM VISUAL ANALYSIS IN FIGS. 7 AND 8

Stage	Conv Layer	Corresponding labels in the figures
Stage 1	Layer 0, Layer 1	(a), (b)
Stage 2	Layer 0, Layer 1	(c), (d)
Stage 3	Layer 0 $\sim$ Layer 8	(e) $\sim$ (m)
Stage 4	Layer 0, Layer 1	(n), (o)

fications. Nonetheless, G-VMamba-B achieved a classification accuracy of 96.89% with a SD of only 4.26%, indicating that as model parameters and data volume increase, the model's performance remains stable.

For the AID dataset, G-VMamba and VMamba exhibited comparable SDs, but G-VMamba achieved superior average classification performance.

In summary, G-VMamba improved performance across most scene classification tasks; however, it was less effective for certain scenes in the NWPU-RESISC45 dataset. Future work should focus on optimizing the balance between feature utilization and generalization across different scenes.

D. Analysis of Class Activation Map

We further analyzed the effectiveness and significance of the ConEM in G-VMamba by visualizing the class activation map (CAM) [61], [62].

1) CAM Comparison of Different Models: We selected FullGrad, which introduces bias contribution, to generate CAM. FullGrad aggregates the contributions from all layers along the path from input to output, focusing on the impact of all model parameters on the input. This generates a comprehensive activation map highlighting which input regions influence the model's decision most. A comparison of CAM visualization results from different models of the experiments is shown in Fig. 6.

The heat map shows the area in the map where the model contributes the most to classification decisions. RSMamba's intersection classification focuses on the edge of the road, which may be disadvantageous if it is a high-speed scene, while our model focuses on the intersection position, which is the most obvious intersection. characteristics. The RSMamba classification of beaches only focuses on seawater. If the proportion of seawater in the beach is small, it may have a negative impact. Sea water + beach is the most important information for identifying beaches.

However, this does not fully reflect the classification performance of the model. In the case of the Beach scene, VMamba directed its attention to the sand, whereas RSMamba focused on the seawater, yet the two models exhibited comparable classification performance. The heat map only indicates where different models focus on when making classification decisions. Based on this, the impact of contour enhancement on the model is shown.

As evident from Fig. 6, compared with VMamba, G-VMamba narrowed the model's attention to areas of color (or luminance) transitions in the image, such as the intersections in the overpass scene, the peripheries of the baseball diamond field, or the shadows cast by the airplane in the airplane scene. These observations underscore the efficacy of the ConEM.

ResNet and ResNeSt demonstrated comparable performance during classification, with both models homing in on objects within the airplane and overpass scenes, like vehicles on the upper deck of the overpass or the airplane's location. When confronted with scenes such as Beach and Baseball diamond, which lack distinct objects, ResNet, ResNeSt, and the Swin transformer tended to concentrate on more coherent areas, emphasizing buildings in the Baseball diamond scene and focusing on waves or sand in the Beach scenes. Compared to the CNN models, the Swin transformer exhibited a more dispersed focus during scene classification, with Rev-ViT showing even greater dispersion.

In the figure, our model always focuses on the locations of the most significant changes from a global perspective. RSMamba's performance is unstable. Sometimes it focuses on too fine-grained locations, and sometimes it focuses on a homogeneous area. Therefore, the summary is that our model has more global advantages.

2) CAM Comparison of Different Feature Layers: To thoroughly demonstrate and analyze the impact of the proposed model on the feature layers, we conducted a CAM comparative analysis on different feature layers during the classification process of the VMamba and G-VMamba models (Figs. 7 and 8).

In Fig. 7, EigenCAM shows the class-independent global activation information. EigenCAM does not rely on category gradients and uses principal component analysis (PCA) to extract the main activation information in the feature map, directly reflecting the activation areas of the image at different feature layers of the model. Fig. 7 explains the global features of the image in 15 blocks across the four stages of the model, illustrating the overall understanding of the image by the two models. Influenced by our proposed ConEM, G-VMamba is more sensitive to image details than VMamba, especially at the fine-scale layer [Fig. 7(a) and (b)].

In Fig. 8, LayerCAM shows a fine-grained activation map based on class gradients. LayerCAM calculates gradient weights pixel by pixel, providing a more detailed visualization. The calculation process is closely related to the gradient of the target class, so the activation area it displays is the area

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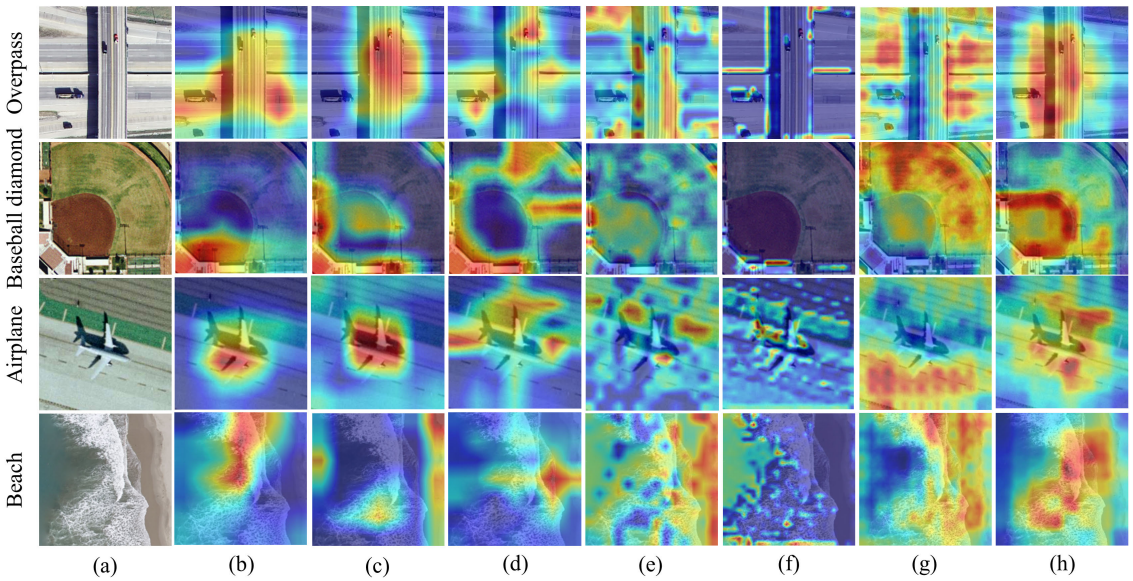


Fig. 6. Comparison of CAM visualizations using different methods on UC-Merced dataset. The colors in this figure represent the contribution of different areas in the input image of a certain category to the model's prediction. Hues closer to red indicate a stronger influence of the area on the classification decision. (a) Original Image, (b) ResNet 50, (c) ResNeSt 50, (d) Swin Transformer, (e) Rev-ViT, (f) RSMamba, (g) VMamba, and (h) G-VMamba.

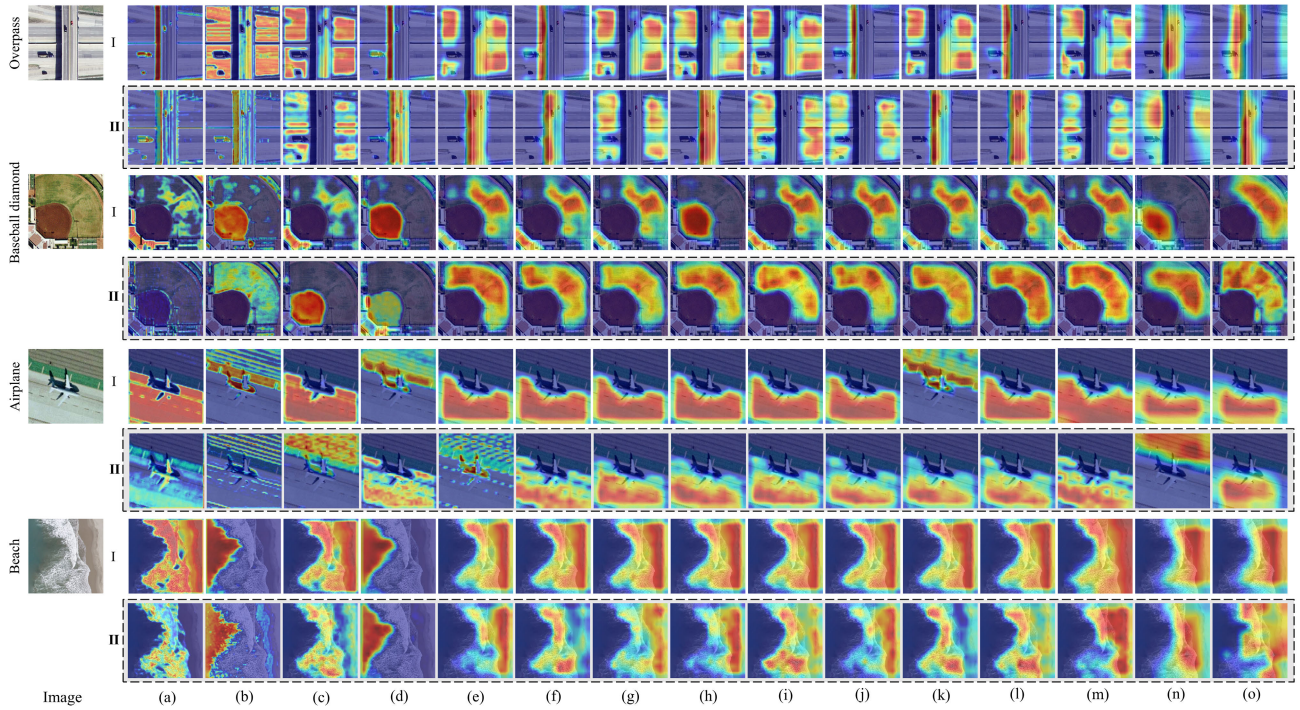


Fig. 7. Feature map visualization using EigenCAM (class-independent). Row I is the VMamba model's feature layers and row II is the G-VMamba model's feature layers. Feature layers (a)–(o) description are given in Table IV.

that is most sensitive to a particular class. It can be seen that between Fig. 8(a) and (o), there are obvious differences in the activation areas of the two models from fine-scale to coarse scale. The proposed G-VMamba focuses on a concentrated range during classification, indicating that it has a stronger ability to distinguish local differences. The best representative is the baseball diamond scene. The feature activation map of Stage 2 in the model from Fig. 8(e)–(m) reflects G-VMamba's ability to focus on brightness changes. The focus on class-related activation areas theoretically positively affects the classification results.

E. Analysis of Feature Fusion

To investigate the impact of varying degrees of luminance gradient feature enhancement during feature fusion in the ConEM, the experimental analyses were conducted using different values for parameter b of (3). As b is increased, it introduces a bias favoring luminance gradient features along with other features, with edge features consistently receiving a greater enhancement than other regions, namely up to $b+1$ times their original values.

As shown in Fig. 9, G-VMamba-T/S exhibit similar trends in response to variations of b. When $b < 1$, the multiplication

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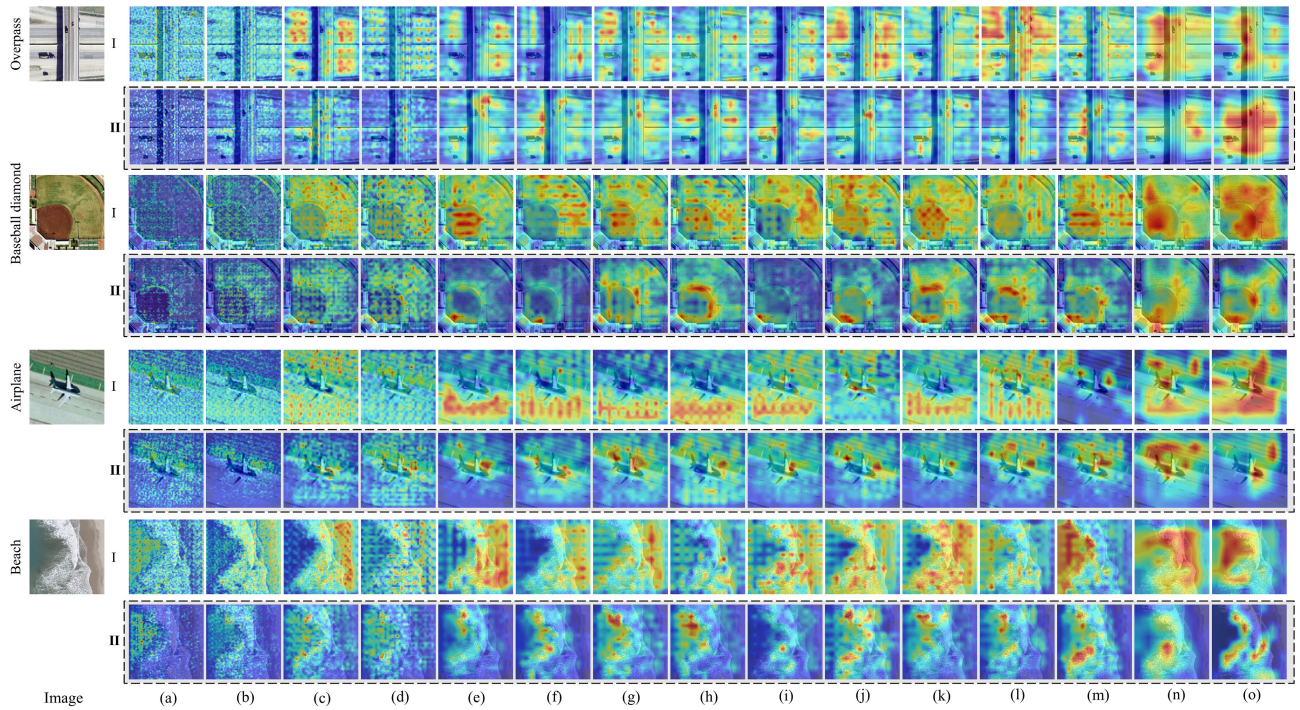


Fig. 8. Feature map visualization using LayerCAM (class-related). Row I is the VMamba model's feature layers and row II is the G-VMamba model's feature layers. Feature layers (a)–(o) description are in Table IV.

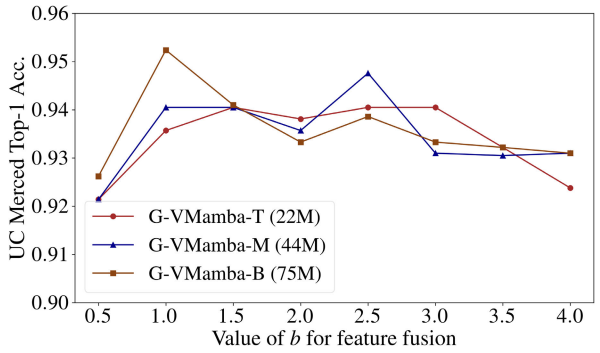


Fig. 9. Model accuracy for different values of b in UC-Merced classification.

of luminance gradient features with the convolutional features results in a weakening of the latter's color and texture representation, which are beneficial for classification; this impacts the model's performance negatively. When $b \in [1, 2.5]$, the classification accuracy improvement for G-VMamba-S remains stable. Beyond $b > 2.5$, the effects on model performance are marginal or even detrimental, indicating that excessive contour information enhancement of features is actually disadvantageous. Adjusting $b \in [0.5, 4.0]$ for G-VMamba-B has a marginal impact, resulting in a difference in classification accuracy of no more than 0.5%. This may be because when the model size increases, the features are extracted more fully and there are more redundant features, so only relying on appropriate contour enhancement does not have a substantial impact on the model. From the above, it is clear that the contribution of luminance gradient features to image classification is, to some extent, limited.

F. Analysis of Features Similarity

To quantitatively validate the effectiveness of VMamba and G-VMamba in distinguishing different classes, we extracted features from three datasets, respectively, and compared the similarities of features of images of different classes. We employ cosine similarity, Spearman's correlation, and T-test as the evaluation methods. The calculated similarity matrix is shown in Fig. 10. The visualization in the three rows above follows the rule that the lighter the color indicates the higher the similarity between the features. The last row of p-values follows the rule that the darker the color, the closer it is to the significance level.

In the first row of Fig. 10, cosine similarity focuses on the direction of the vectors. The closer the value is to 1, the more similar the features are. When comparing the differences between VMamba and G-VMamba, we find that when using the VMamba model for prediction, the cosine similarity between different classes is very high, with the average similarity of features across three datasets reaching 0.90, 0.97, and 0.97, respectively. When using G-VMamba, the average similarity of features in the UC Merced dataset decreases to 0.55, indicating that the differences between different classes are increased, and correspondingly, the colors in the similarity matrix become more diverse. For the AID dataset and the NWPU-RESISC45 dataset, the average similarities are 0.85 and 0.81, respectively, with a relatively smaller overall decrease. However, the changes in the similarity matrices are noticeable, with the absolute differences between some classes being increased. For some classes, G-VMamba performs exceptionally well, such as Overpass, Beach, and River. These scenes have distinct outlines. However, this is not the case for other classes, such as Dense residential and Desert.

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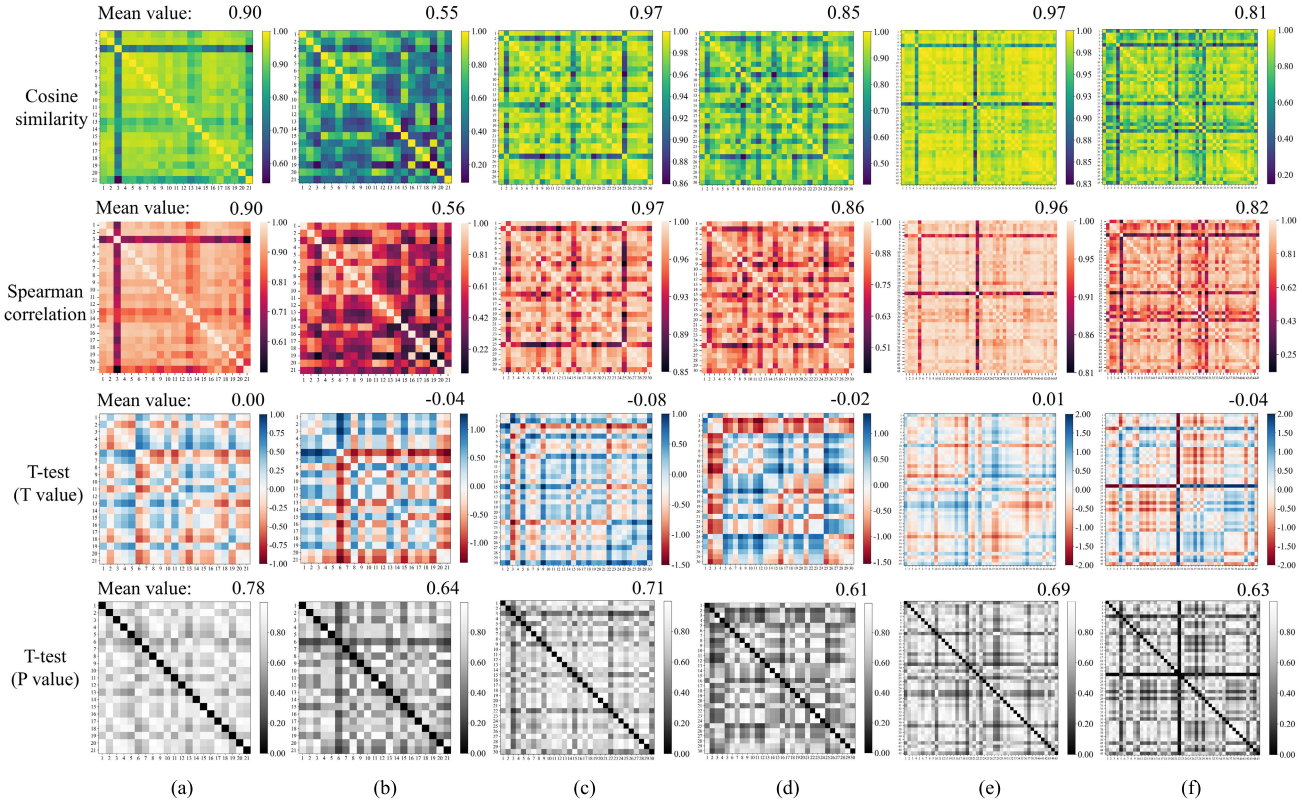


Fig. 10. Similarity analysis of image features for different classes extracted using VMamba and G-VMamba. (a) and (b) UC Merced [58], (c) and (d) AID [60], and (e) and (f) NWPU-RESISC45 [59]. Note: The visualization in the three rows above follows the rule that the lighter the color indicates the higher the similarity between the features. The last line of p-values follows the rule that the darker the color, the closer it is to the significance level. *UC Merced dataset classes are*: 1—Agricultural, 2—Airplane, 3—Baseball diamond, 4—Beach, 5—Buildings, 6—Chaparral, 7—Dense residential, 8—Forest, 9—Freeway, 10—Golf course, 11—Harbor, 12—Intersection, 13—Medium residential, 14—Mobile home park, 15—Overpass, 16—Parking lot, 17—River, 18—Runway, 19—Sparse residential, 20—Storage tanks, and 21—Tennis court. *AID dataset classes are*: 1—Airport, 2—Bare Land, 3—Baseball field, 4—Beach, 5—Bridge, 6—Center, 7—Church, 8—Commercial, 9—Dense residential, 10—Desert, 11—Farmland, 12—Forest, 13—Industrial, 14—Meadow, 15—Medium residential, 16—Mountain, 17—Park, 18—Parking, 19—Playground, 20—Pond, 21—Port, 22—Railway station, 23—Resort, 24—River, 25—School, 26—Sparse residential, 27—Square, 28—Stadium, 29—Storage tanks, and 30—Viaduct. *NWPU-RESISC45 dataset classes are*: 1—Airplane, 2—Airport, 3—Baseball diamond, 4—Basketball court, 5—Beach, 6—Bridge, 7—Chaparral, 8—Church, 9—Circular farmland, 10—Cloud, 11—Commercial area, 12—Dense residential, 13—Desert, 14—Forest, 15—Freeway, 16—Golf course, 17—Ground track field, 18—Harbor, 19—Industrial area, 20—Intersection, 21—Island, 22—Lake, 23—Meadow, 24—Medium residential, 25—Mobile home park, 26—Mountain, 27—Overpass, 28—Palace, 29—Parking lot, 30—Railway, 31—Railway station, 32—Rectangular farmland, 33—River, 34—Roundabout, 35—Runway, 36—Sea ice, 37—Ship, 38—Snowberg, 39—Sparse residential, 40—Stadium, 41—Storage tank, 42—Tennis court, 43—Terrace, 44—Thermal power station, and 45—Wetland.

These scenes have dense textures or appear uniform, making it difficult to distinguish classes based on outlines. In such cases, the model can only rely on color, texture information, and other advanced features.

In the second row of Fig. 10, Spearman's correlation focuses on the rank order. The closer the absolute value is to 0, the more similar the features are. The matrices of cosine similarity in the first row and Spearman's correlation in the second row of Fig. 10 are very similar, indicating that the features have consistent structure and relationships across different dimensions. comparing the color distributions of the feature similarity matrices for the same data (comparing Fig. 10(c) and (d), or (e) and (f)), the relative differences between VMamba and G-VMamba are close. This indicates that G-VMamba, building upon VMamba, maintains the original feature extraction capabilities while increasing the differences between some data features, thereby leading to a stable improvement in model performance.

The third row of Fig. 10 shows the T-test-based similarity matrix. The t-value directly reflects the overall mean difference; the closer it is to 0, the less difference there is. Because the data was normalized during input, the fluctuation range of the feature values is very small, and the mean and variance of the features have very weak discriminability. We perform a logarithmic transformation ($X' = e^X$) on all features during the T-test to amplify the differences for visualization. The T-test shows that G-VMamba and VMamba have different responses to features of different categories, but the differences are not significant. The feature differences of classes in the UC Merced and AID datasets are less than 1, and the differences in the NWPU-RESISC45 datasets are less than 2. The G-VMamba model alters the mean level of feature values for individual image categories, most noticeably for the "22—Lake" in the NWPU-RESISC45 dataset. However, in the fourth row of Fig. 10, the p-values for the T-test are larger overall. Taking 0.05 as the significance level, the conclusion is that there are no significant differences between the features.

In summary, combined with the conclusions of CAM visualization analysis, the analysis results of cosine similarity and Spearman correlation further verify that the ConEM module

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of G-VMamba changes the spatial distribution of feature perception. T-test verifies that the overall mean and variance of all features are not much different.

V. Conclusion

In this article, the contour-enhanced Mamba-based model G-VMamba is presented. Its main novelty is that at each network layer, hand-crafted gradient features and multidimensional convolutional features are extracted separately before being fused for image classification. This approach strengthens the model's sensitivity to color (or luminance) variations while imposing constraints on low-level features in deeper networks. The model's effectiveness is demonstrated through experiments on high-resolution RS image classification. While the improved model demonstrated improved performance compared to other state-of-the-art models, the limitation of the luminance gradient enhancement being applied solely during the forward propagation phase to input features resulted in less pronounced performance improvements in large-scale datasets.

Furthermore, through a visual analysis of the CAM and feature similarity, it is evident that enhancing the model's responsiveness to regions with significant color (or luminance) changes contributes meaningfully to increase the feature differences of different ground feature types and its improved classification performance. In the future, our focus will lie on prudently harnessing the advantages of hand-crafted gradient features and exploring integration schemes that combine luminance variation, color, and texture characteristics, as these are considered of significant importance to the study of explainable DL.

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